

System Level Tests - LightSpeed 4.X

1 Module Overview

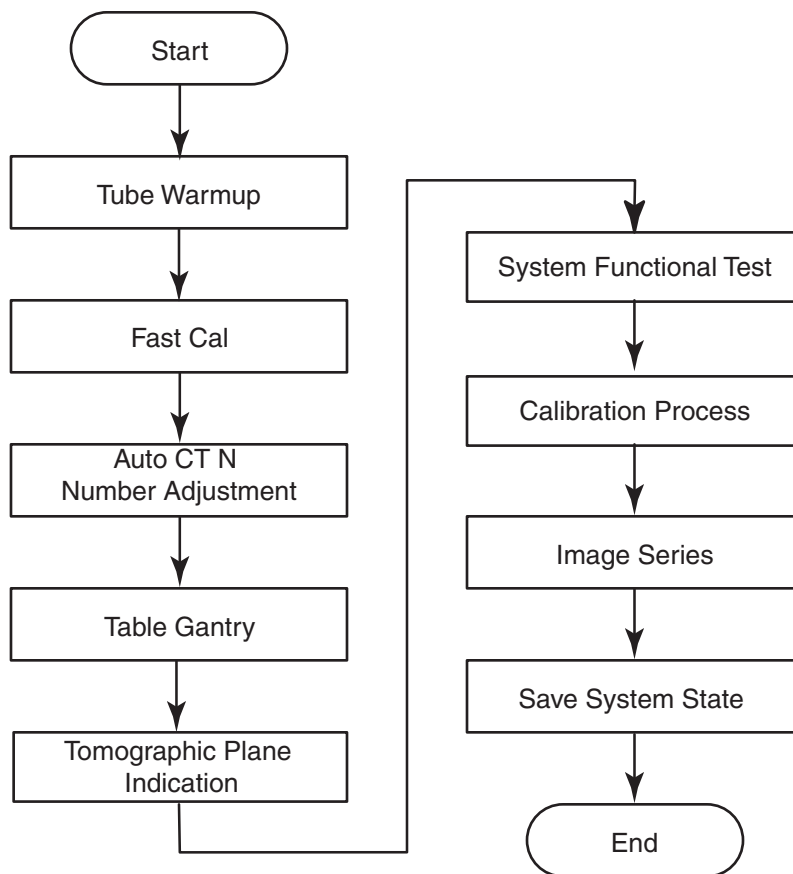
The System Level Tests for the LightSpeed 4.X 16 slice include the following sections:

- [Introduction](#)
- [Autovoice/Intercom Check](#)
- [Tube Warm Up and Fast Cal](#)
- [Auto CT N Number Adjustment](#)
- [Image Series Checks Overview](#)
- [48cm Phantom Image Series Verification](#)
- [35cm QA Phantom Image Series Verification](#)
- [System Functional Test](#)
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2 Introduction

This section contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

Figure 1: System Test Overview

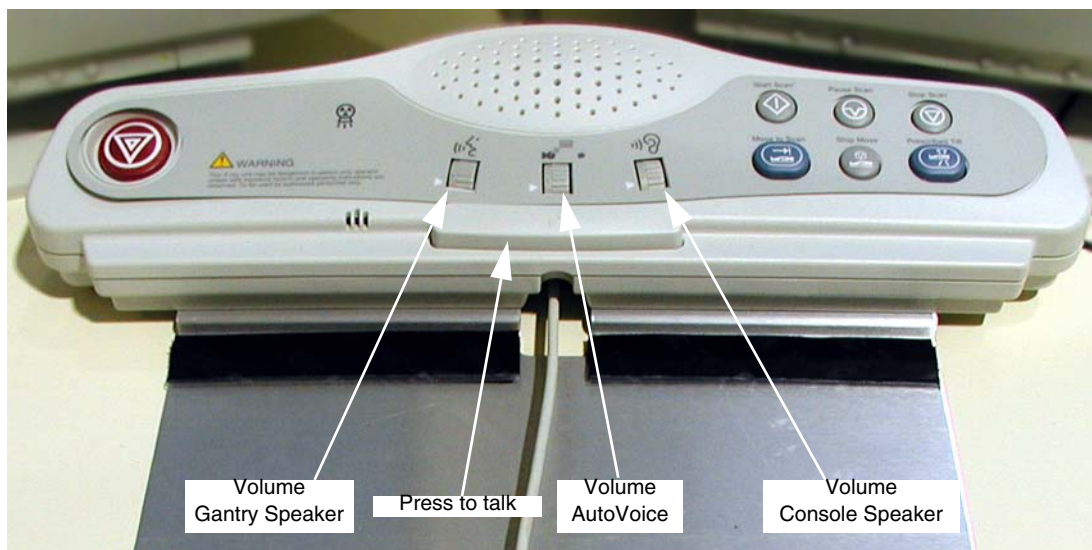


3 Autovoice/Intercom Check

3.1 Gantry Speaker

To adjust the gantry speaker volume, adjust the left-most volume thumb wheel while speaking into the console microphone (press the bar on the SCIM — see [Figure 2](#)).

Figure 2: SCIM Volume Controls

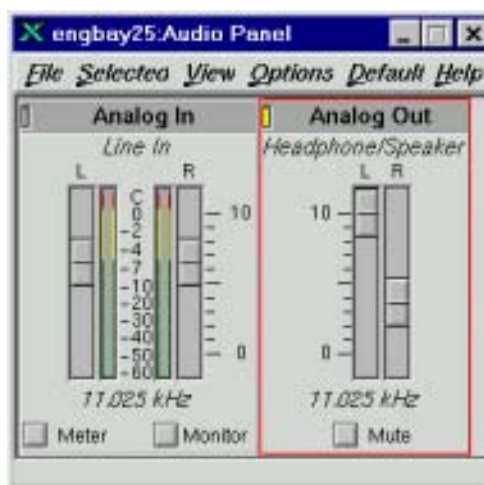


3.2 Console Speaker

To adjust the console speaker volume:

- 1.) Click on the Autovoice volume control from the tool chest.
- 2.) Click on the Audio Panel GUI icon to open it.
- 3.) Leave all audio panel sliders at their default positions, as shown in [Figure 3](#).

Figure 3: Audio Panel



- 4.) Select FILE-SAVE when finished to retain your settings.

3.3 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click on the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click on the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel, while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5, as necessary to achieve satisfactory volume.
- 7.) Select DONE, then Select QUIT.

NOTE: If a satisfactory volume can not be achieved, refer to the *LightSpeed 4.X System Service Manual* and review the factory setup for the:

- console intercom pots
- gantry intercom pots

4 Tube Warm Up and Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

NOTE: Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

5 Auto CT N Number Adjustment

NOTE: If it has been more than 30 minutes since calibration scans, rerun FASTCAL (approx 15 min). N Number Adjustment needs the tube to be warm, or CT numbers will change.

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Place the QA Phantom in the holder.
- 2.) Center the Phantom
- 3.) From the application desktop, select SCANNER UTILITIES.
- 4.) Select AUTO CT NUMBER ADJUSTMENT.
- 5.) Follow directions on the screen.
 - a.) When centering the phantom: (refer to
 - * Centering may be easier to perform from the gantry right side.
 - * Center on the water portion of the phantom (not over the internal block).
 - b.) The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

Initial: _____

6 Image Series Checks Overview



NOTICE Run **ALL Image Series Tests** in Auto Mode. Manual procedures are provided in this section as **REFERENCE ONLY**.

6.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

NOTE: Consider *any* image series scan that does not meet specifications as failing.

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

6.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

6.3 Term Definitions and Screens

Xc - MEAN CT NUMBER FOR THE SPECIFIED CENTER COORDINATES OF THE PHANTOM IMAGE.

AvXc - AVERAGE MEAN CT NUMBER FOR THE CENTER OF THE PHANTOM IMAGE: AVERAGE THE MEAN CT VALUE FOR ALL SPECIFIED CENTER COORDINATES OF ALL SLICES IN AN EXAM.

Xo - MEAN CT NUMBER FOR THE OUTSIDE OF THE PHANTOM IMAGE: AVERAGE THE MEAN CT VALUE FOR ALL SPECIFIED OUTSIDE COORDINATES OF ONE SLICE.

AvXo - AVERAGE OUTSIDE MEAN CT NUMBER FOR THE NUMBER OF SLICES IN AN EXAM.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - AVERAGE IMAGE NOISE (STANDARD DEVIATION) FOR THE OUTSIDE OF A PHANTOM: AVERAGE OF ALL OUTSIDE COORDINATES OF ALL THE SLICES IN AN EXAM.

Figure 4: Image Analysis Tool User Interface - Auto 1x Test Pull Down Menu

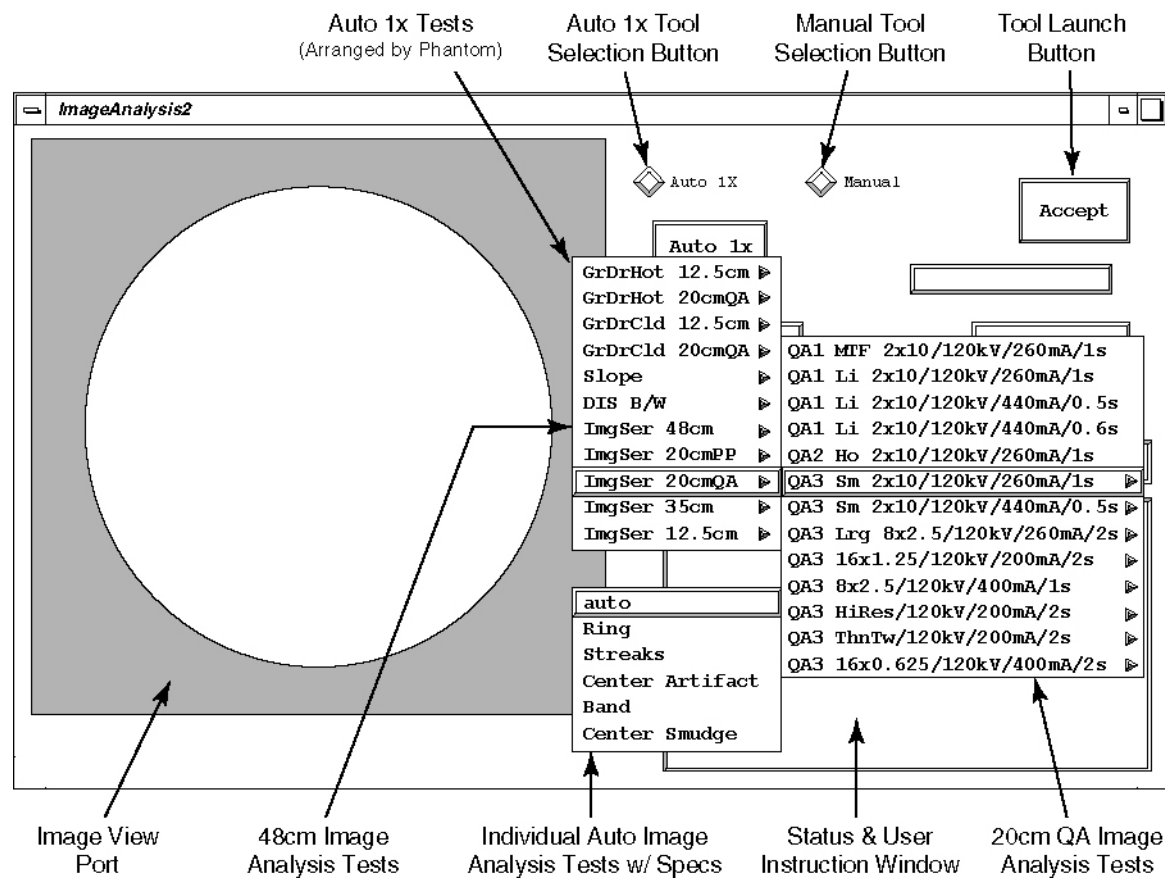
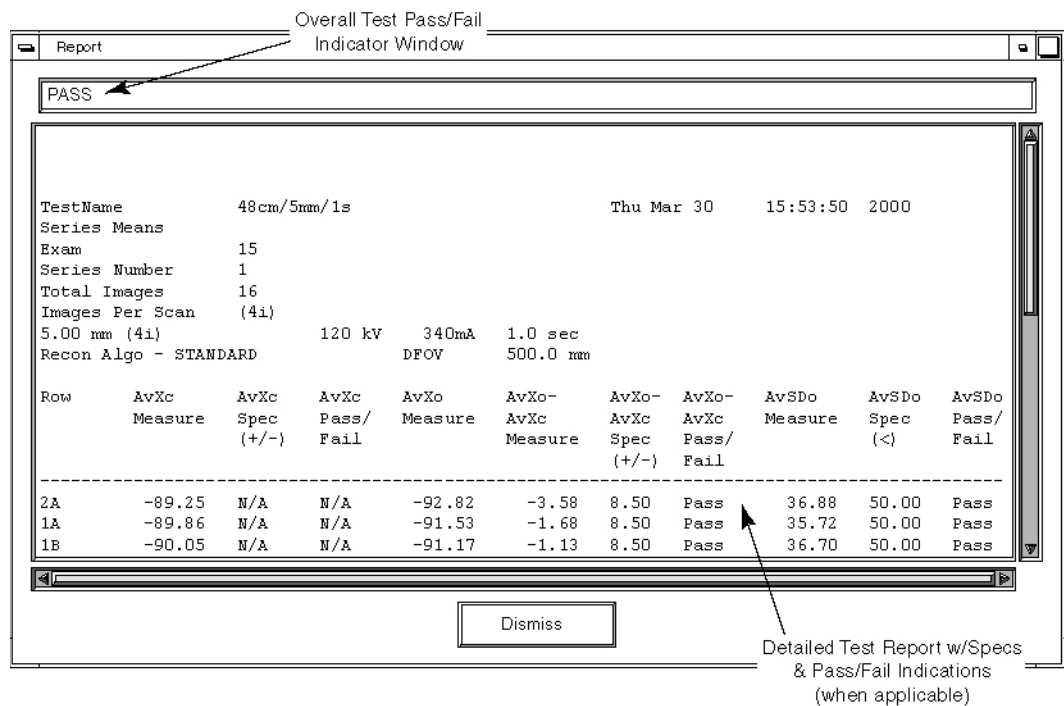


Figure 5: Image Analysis Tool User Interface - Test Results Report Window



7 48cm Phantom Image Series Verification

7.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a single, 64 image, 48cm Phantom series.

Manual Scan Protocol Setup

Refer to [Table 1](#) and set-up an Axial scan with the parameters shown.

Table 1: 48cm Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 16x1.25/120kV /400mA/2s	Axial Full 2.0 sec	19.375	59.375	64	1.2516i	0.00	S0.0	Large	120	400	8.0 sec.	50.0	Std

Auto Scan Protocol Setup

- a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the two listed Patient Information fields:
 - * Patient ID: **Service**
 - * Name: **48cm Phantom Image Series**
 - c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
 - d.) On the Service Protocol Selection window, click on MANUFACTURING.
 - e.) On the Service Category Protocol List, click on 45.7 IMGSER 48CM MDAS_16_16.
 - f.) On the ExamRx protocol parameter display, skip to the 3rd Series (Series Description: 48 16x1.25/120kV/400mA/2s).
 - g.) Set internal Landmark.
- 5.) Acquire a single, 64 image, 4 scan, image series of the 48 cm Phantom.

7.2 Image Performance Verification

7.2.1 Series Means (AvXo-AvXc and AvSDo) & Center Smudge

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on the IMAGE QUALITY icon.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.
- NOTE: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.
- 2.) Using the AUTO 1X Tool on the Image Analysis screen, generate a Series Means and Center Smudge report for the 1st series of the 48cm Phantom acquisition.
 - a.) From the Service Browser window, select the 48cm Phantom series acquired in Section 7.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button.
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S from the test pull-down menu, and continue to slide cursor to the right to select AUTO.
 - d.) Click on the ACCEPT button.
 - 3.) Review the Image Analysis Report data and record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 2](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - b.) Verify the scan parameter values displayed in the report match those in [Table 1](#).
 - c.) Record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 2](#).

7.2.2 Band and Streak Artifact

- 1.) Review the 48cm Phantom images acquired in Section 7.1 for Bands and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Browser window, select the 48cm Phantom series acquired in Section 7.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the 64 images in the series, analyze images for any Band or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Band or Streak artifact.
- 2.) Verify any images noted above (step 1e) meet 48cm Phantom Band or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 48cm Phantom series acquired in Section 7.1 that appeared to have a Band or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S from the test pull-down menu, and continue to slide cursor to the right to select BAND or STREAK.

- d.) Click on the ACCEPT button.
- NOTE: If required, adjust Window/Level using the center mouse button.
- e.) Position and size the Band or Streak ROI using the left cursor button.
- NOTE: To ensure specifications are properly applied, do not adjust the Reference ROI's for the Band and Streak artifact tests. The Reference ROI's are adjustable after the 1st Accept Modification click.
- f.) Click on ACCEPT MODIFICATION twice, to generate a report.
- g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
- h.) Repeat steps b through g, for each image noted in Step 1e.

Box Size =1600 mm² to 2500 mm²

45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)

45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions:Box 1= 0 mm x 0 mm

Box 2 = 0 mm x -190 mm

Box 3 = 190 mm x 0 mm

Box 4 = 0 mm x 190 mm

Box 5 = -190 mm x 0 mm

Table 2: 48cm Phantom CT# Brightness Uniformity, Noise & Center Smudge Row Performance Data Sheet

Row	Images	Brightness Uniformity (AvXo - AvXc)	Noise (AvSDo)	Center Smudge Row Values	Comments
8A	1, 17, 33, 49				
7A	2, 18, 34, 50				
6A	3, 19, 35, 51				
5A	4, 20, 36, 52				
4A	5, 21, 37, 53				
3A	6, 22, 38, 54				
2A	7, 23, 39, 55				
1A	8, 24, 40, 56				
1B	9, 25, 41, 57				
2B	10, 26, 42, 58				
3B	11, 27, 43, 59				
4B	12, 28, 44, 60				
5B	13, 29, 45, 61				
6B	14, 30, 46, 62				
7B	15, 31, 47, 63				
8B	16, 32, 48, 64				
Specifications		< +/- 17.0	< 50	< +/- 14	

7.3 Failure Recommended Actions

7.3.1 Series Means (AvXo - AvXc or AvSDo) Failure Recovery

7.3.1.1 Specifications

Each Row of the series must pass 48cm Series Means (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 17.0
- AvSDo: < 50.0

7.3.1.2 Recommended Recovery

- 1.) Repeat Detailed Cal for the 48cm Phantom.
- 2.) Repeat Sections 7.1 & 7.2, to verify Image Performance.

7.3.2 Center Smudge Failure Recovery

7.3.2.1 Specifications

80% the Rows must pass Center Smudge specifications when re-scanning:
Center Smudge:Smudge Factor < +/- 14.0

7.3.2.2 Recommended Recovery

- 1.) If one image of one Row fails (Row fails once):
 - Re-scan the 48cm Phantom acquiring two series of 64 images using the same protocol settings as listed in [Table 1](#) (on [page 9](#)).
 - Analyze the two 64 re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 128 re-scanned images do not fail Series Means (AvXo - AvXc and AvSDo) or Center Smudge.
- 2.) If more than one image of one Row fails (Row fails more than once) or multiple rows are failing:
 - Repeat Detailed Cal for the 48cm Phantom.
 - Re-scan the 48cm Phantom acquiring three series of 64 images using the same protocol settings as listed in [Table 1](#) (on [page 9](#)).
 - Analyze the three re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 192 re-scanned images do not fail Series Means (AvXo - AvXc and AvSDo) or Center Smudge.

7.3.3 Band or Streak Artifact Failure Recovery

7.3.3.1 Specifications

- Band Artifact: Band Factor< +/- 8.0
- Streak Artifact: Streak Factor< +/- 4.0

7.3.3.2 Recommended Recovery

- 1.) Troubleshoot Band or Streak artifact failures, as described in the *System Service Manual*.
- 2.) Repeat Detailed Cal for the 48cm Phantom.
- 3.) Repeat Sections 7.1 & 7.2, to verify Image Performance.

8 35cm QA Phantom Image Series Verification

8.1 Acquiring the 35cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 35cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 35cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan three 35cm QA Phantom image series with a Recon of the 1st series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 3](#) and set-up an Axial scan with the parameters shown.

Table 3: 35cm QA Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA1 MTF 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	I5.00	S5.00	4	10.02i	0.00	S0.0	Small	120	260	2.0 sec.	25.0	Std
Recon 2: Q	Axial Full 1.0 sec.	I5.00	S5.00	4	10.02i	0.00	S0.0	Small	120	260	2.0 sec.	15.0	Bone
QA2 Ho 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	S40.00	S50.00	8	10.02i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std
QA3 Sm 2x10/120kV/ 260mA/1s	Axial Full 1.0 sec.	S55.00	S65.00	8	10.02i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **35cm QA Phantom Image Series**
- c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
- d.) On the Service Protocol Selection window, click on MANUFACTURING.
- e.) On the Service Category Protocol List, click on 45.9 IMGSER 35CMQA MDAS_16_16.
- f.) On the ExamRx protocol parameter display, select the 1st Series (Series Description: QA1 MTF 2x10/120kV/260mA/1s).
- g.) Set internal Landmark.
- 5.) Acquire the 1st 35cm QA Phantom image series and one Recon series (MTF and Contrast, Visible Lines).
- 6.) Acquire the 3rd 35cm QA Phantom image series (Series Description: QA2 Ho 2x10/120kV/260mA/1s).
- 7.) Acquire the 4th 35cm QA Phantom image series (Series Description: QA3 Sm 2x10/120kV/260mA/1s).

8.2 35cm QA Phantom Image Series Image Performance Verification

1st QA Phantom Image Series (4 Image MTF Average and Contrast Scale) Image Performance Verification

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on IMAGE QUALITY TESTS.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.

NOTE:

During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.

- 2.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#1 MTF and QA#1 Contrast Scale report for the 1st series of the 35cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st series of the 35cm QA Phantom series acquired in section 8.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button.
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 35CMQA test from the pull-down menu, slide cursor to the right to select QA1 MTF 2X10/120KV/260MA/1S from the test pull-down menu.
 - d.) Click on the ACCEPT button.
- 3.) Review the Image Analysis Report data and record the Per Image MTF, 4 Image MTF Average, and Per Image Contrast Scale data for each row in Table 10-7.
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3](#) (on [page 13](#)) for the 1st 35cm QA Image Series.
 - c.) Record Per Image MTF and Per Image Contrast Scale data for each image in [Table 4](#).
 - d.) Record the 4 Image MTF Average in [Table 4](#).

Table 4: 35cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (MTF and Contrast Scale)

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		N.A.		
2		N.A.		
3		N.A.		
4		N.A.		
Specifications	N.A.	0.65 to 1.0	110.0 to 130.0	N.A.

8.3 2nd QA Phantom Image Series (4 Bone Retro Image Visible Lines) Image Perf. Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#1 Lines report for the 2nd series of the 35cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 2nd series of the 35cm QA Phantom series acquired in section 8.1 (on [page 13](#)).
 - b.) Click on the AUTO 1X rectangular button, select the IMGSER 35CMQA test from the pull-down menu, and slide cursor to the right to select QA1 LI 2X10/120KV/260MA/1S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Lines pop-up window.
 - d.) On the Visible Lines pop-up window, click on the VISIBLE LINE button.
- NOTE: The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.
- e.) From the Visible Line pull-down menu, select the letter (A through F) that matches the smallest line pair pattern that you can discern in the Image View Port.
- f.) On the Visible Lines pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#1 Visible Lines data for the 1st image in [Table 5](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3](#) (on [page 13](#)) for the 2nd 35cm QA Image Series.
 - c.) Record the Per Image QA#1 Visible Lines data for the 1st image in [Table 5](#).
- 3.) Repeat Steps 1 and 2, for each of the remaining three images in the 2nd 35cm QA Image Series.

Table 5: 35cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (Visible Lines)

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	N.A.

8.4 3rd QA Phantom Image Series (4 Image Visible Holes) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#2 Holes report for the 3rd series of the 35cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 3rd series of the 35cm QA Phantom series acquired in section 8.1.
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 35CMA from the pull-down menu, and slide cursor to the right to select QA2 HO 2X10/120KV/260MA/1S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Hole pop-up window.
 - d.) On the Visible Hole pop-up window, click on the VISIBLE HOLE button.
- NOTE: The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.
- e.) From the Visible Hole pull-down menu, select the number (1 through 5) that matches the smallest hole that you can discern in the pattern centered in the image displayed in the Image View Port.
- f.) On the Visible Hole pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - b.) Verify the scan parameter values displayed in the report match those in [Table 3](#) for the 3rd 35cm QA Image Series.
 - c.) Record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6](#).
- 3.) Repeat Steps 1 and 2 for the 3rd, 5th, and 7th images in the 3rd 35cm QA Image Series.

Table 6: 35cm QA#2 Phantom Low Contrast Detectability Image Performance (Visible Holes)

Image	Visible Holes Viewable at Window 20	Contrast Factor	Comments
1			
3			
5			
7			
Specifications	See Table 6-9	2.0 to 12.0	N.A.

Table 7: 35cm QA#2 Phantom Visible Hole Specifications

Contrast Factor Range (Box 1 Means - Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit *	Upper Limit *	
2.00 to 3.99	2	5	7.5 mm
4.00 to 7.99	3	5	5.0 mm
8.00 to 12.00	4	5	3.0 mm
* Required Number of Visible Holes depends on the Contrast Factor			

8.5 4th QA Phantom Image Series (QA#3 Small) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#3 Small report for the 4th series of the 35cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 4th series of the 35cm QA Phantom series acquired in section 8.1 (on [page 13](#)).
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 35CMQA from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select AUTO.
 - c.) Click on the ACCEPT button.
- 2.) Review the Image Analysis Report data and record the QA3 Small data for the 4th 35cm QA Image Series in [Table 8](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - b.) Verify that the scan parameter values displayed in the report match those in [Table 3](#) (on [page 13](#)) for the 4th 35cm QA Image Series.
 - c.) Record the QA3 Small data (Row 2A1A and Row 2B1B Series Means AvXc and AvXo - AvXc), QA3 (AvSDc), and the Center Smudge Row data in [Table 8](#).

Box Size = 196 mm² to 256 mm²

15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)

31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm

Box 2 = 0 mm x -80 mm

Box 3 = 80 mm x 0 mm

Box 4 = 0 mm x 80 mm

Box 5 = -80 mm x 0 mm

Table 8: 35cm QA#3 Phantom CT#, Brightness Uniformity, Center Noise & Center Smudge Row Performance Data Sheet

Row	Images	AvXc	AvXo	AvXo - AvXc	AvSDo	AvSDc	Center Smudge Row Value	Comments
2A1A	1, 3, 5, 7				N.A.			
1B2B	2, 4, 6, 8				N.A.			
Specifications		+3.0 to -3.0	N.A.	+3.0 to -3.0	N.A.	+ 2.9 to +3.5	+2.2 to -2.2	

- 3.) Review the 4th 35cm QA Phantom images acquired in Section 8.1 for Rings, Bands, and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS DESKTOP.
 - b.) From the Image Works Browser window, select the 4th 35cm QA Phantom series acquired in section 8.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the eight images in the series, analyze images for any Ring, Band, or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Ring, Band, or Streak artifact.

- 4.) Verify any images noted above (step 3e) meet 35cm QA Phantom Ring, Band, or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 4th 35cm QA Phantom series acquired in section 8.1 that appeared to have a Ring, Band, or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 35CMQA test from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select RING, BAND, or STREAK.
 - d.) Click on the ACCEPT button.
- NOTE: If required, adjust Window/Level using the center mouse button.
- e.) Position and size the Ring, Band, or Streak ROI, using the left cursor button.
- NOTE: To ensure specifications are properly applied, do not adjust the Reference ROIs for the Ring, Band, and Streak artifact tests. The Reference ROIs are adjustable after the 1st Accept Modification click.
- f.) Click on ACCEPT MODIFICATION twice to generate a report.
 - g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - h.) Repeat steps b through g for each image noted in Step 3e.

8.6 Failure Recommended Actions

8.6.1 1st Image Series (4 Image MTF Average) Failure Recovery

8.6.1.1 Specifications

The 4 Image MTF Average must pass specifications:

4 Image MTF Average: 0.65 to 1.00

8.6.1.2 Recommended Recovery

- 1.) Repeat Detailed Cal for the 35cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.2 2nd Image Series (Visible Lines) Failure Recovery

8.6.2.1 Specifications

The Largest 5 line pair patterns (coded F, E, D, C, and B) must be visible for each of the four images in this series.

8.6.2.2 Recommended Recovery

NOTE: The most common failure for this test is that the phantom has air bubbles that are obscuring the line pair patterns.

- 1.) Carefully inspect the 35cm QA Phantom for air bubbles. If required, refill the phantom with water, to eliminate all air bubbles.
- 2.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.3 3rd Image Series (Visible Holes) Failure Recovery

8.6.3.1 Specifications

The number of visible holes in this image series must pass specifications as follows:

- Contrast Factor = 2.0 to 3.99:
Minimum of 2 holes must be visible (10mm & 7.5mm)
- Contrast Factor = 4.0 to 7.99:
Minimum of 3 holes must be visible (10mm, 7.5mm & 5mm)
- Contrast Factor = 8.0 to 12.0:
Minimum of 4 holes must be visible (10mm, 7.5mm, 5mm, & 3mm)

8.6.3.2 Recommended Recovery

- 1.) Repeat Detailed Cal for the 35cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.4 4th Image Series Failure Recovery

8.6.4.1 Series Means Specifications

Both Rows (2A1A, 1B2B) of the series must pass QA#3 Series Means specifications:

- AvXc: < +/- 3.0
- AvXo - AvXc: < +/- 3.0

8.6.4.2 Series Means Recommended Recovery

- 1.) Repeat Detailed Cal for the 35cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.4.3 QA#3 (AvSDc) Specifications

Both Rows (2A1A, 1B2B) of the series must pass QA#3 AvSDc specifications:

- AvSDc (< 5K Scans):2.90 to 3.50
- AvSDc (> 5K Scans):2.80 to 3.60

8.6.4.4 QA#3 (AvSDc) Recommended Recovery

- 1.) Repeat Detailed Cal for the 35cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.4.5 Center Artifact or Center Smudge Specifications

All eight images of the series must pass 35cm QA#3 Center Artifact and Center Smudge specifications:

- Center Artifact Factor: +/- 3.5
- Center Smudge Factor: +/- 2.2

8.6.4.6 Center Artifact or Center Smudge Recommended Recovery

- 1.) Repeat Detailed Cal for the 35cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

8.6.4.7 Ring, Band, or Streak Artifact Failure Recovery Specifications

- Ring Artifact: Ring Factor $< \pm 4.8$
- Band Artifact: Band Factor ± 2.8 from 0cm to 8.5cm
 $< \pm 2.8 > 8.5\text{cm}$
- Streak Artifact: Streak Factor $< \pm 4.0$

8.6.4.8 Ring, Band, or Streak Artifact Failure Recovery Recommended Recovery

- 1.) Troubleshoot Ring, Band or Streak artifact failures as described in the Service Manual.
- 2.) Repeat Detailed Cal for the 35cm QA Phantom.
- 3.) Repeat Sections 8.1 & 8.2, to verify Image Performance.

9 System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.

NOTE:

Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.

-or-

Manually select the scan parameters specified in [Table 9](#).

Table 9: System Functional Test Scan Parameters

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot
Air	120	50		16 x 1.25	1.0sec			
Air	120	50		8 x 2.5	1.0sec			

- 3.) Review all air scans for presence of any artifacts (rings, bands, streaks, etc.).
- 4.) If any artifacts are present, review the MSD Plots for each AIR scan, using Diagnostic Data Analysis.
- 5.) Complete the scans.

Initial: _____

10 Save System State

Use the following commands to create the System State disk (DVD or MOD).

- 1.) Load the archive medium (DVD or MOD) into the archive drive, in the SCSI tower.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 3.) Click on the PM icon.



- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.

- 8.) Remove the archive medium (DVD or MOD) from the drive.